

Employment-Unemployment Indicator Estimation

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PROTOTYPE LEARNING MATERIAL NOTICE

This learning module is compiled from curriculum standards for role-aligned training demonstration.

Table of Contents

1. Introduction
2. Key Concepts
3. Estimation Methodology
4. Administrative Data Sources for Employment Estimation
5. Interpretation and Common Pitfalls
6. Practical Example: Computing State-level LFPR
7. Data Quality Considerations
8. Key Learning Points
9. Learning Outcomes

1. Introduction

Employment and unemployment indicators are among the most closely watched economic statistics, directly informing labour policy, social protection design, and macroeconomic management. In India, the primary source for these indicators is the Periodic Labour Force Survey (PLFS), supplemented by administrative data from EPFO, ESIC, NPS, and payroll-based estimates.

This course focuses on the estimation methodology for employment-unemployment indicators: how they are computed from survey data, how they are validated, and how they are interpreted in the Indian context. It also covers the internationally recommended definitions from ILO's Resolution on Statistics of Work, Employment and Labour Underutilisation (2013 ICLS).

2. Key Concepts

2.1 ILO / ICLS 2013 Definitions

The 19th International Conference of Labour Statisticians (ICLS, 2013) redefined key concepts:

****Employment****: Persons who, during the reference period, worked for at least one hour in exchange for pay or profit, or worked in a job regardless of any pay (contributing family worker).

****Unemployment****: Persons who: (a) are not in employment, (b) are available for employment, and (c) are seeking employment. All three conditions must be met simultaneously.

****Labour underutilisation****: A broader concept covering unemployment + time-related underemployment + potential labour force (discouraged workers).

2.2 Usual Activity Status (UAS)

Determined on the basis of the major time criterion over a 365-day reference period. Categories:

- Workers (self-employed: own account, employer, contributing family worker; salaried/regular; casual wage)
- Unemployed (seeking and available)
- Out of labour force (students, domestic duties, pensioners, disabled)

2.3 Employment Status Categories

****Self-employed**:** Own-account workers (work alone), Employers (hire others), Contributing family workers (unpaid work in family enterprise).

****Regular/Salaried employees**:** Work for others on a regular basis and receive salary/wage.

****Casual wage labourers**:** Work on a day-to-day or casual basis; do not have regular/long-term work contract.

2.4 Formal vs. Informal Employment

Informal employment encompasses all jobs lacking basic social or legal protections or employment benefits-whether in formal or informal sector enterprises. Estimated using PLFS variables on job contract type, social security coverage, and written job contract. India has a very high share (~90%) of informal employment.

3. Estimation Methodology

3.1 Step 1: Weight Assignment

Each sampled household is assigned a design weight = inverse of the probability of selection. Weights are calibrated to known population totals (from Census/population projections) by state, sector (rural/urban), age group, and sex. This corrects for differential non-response and improves precision.

3.2 Step 2: Activity Classification

Each person 15+ years is classified into activity status categories (worker/unemployed/OOLF) based on:

- Usual Principal Activity (UPA): Major time in 365 days
- Usual Subsidiary Activity (USA): Minor time in 365 days
- Current Weekly Status (CWS): Activity in past 7 days
- Current Daily Status (CDS): Activity each day of reference week

3.3 Step 3: Indicator Computation

Weighted estimates of employed, unemployed, and labour force are computed by summing person-weights across relevant categories:

$$\text{Estimated Employed} = \sum (\text{weight}_i \times \text{employed_indicator}_i)$$

$$\text{LFPR} = (\text{Estimated Labour Force} / \text{Estimated Population 15+}) \times 100$$

$$\text{WPR} = (\text{Estimated Employed} / \text{Estimated Population 15+}) \times 100$$

$$\text{UR} = (\text{Estimated Unemployed} / \text{Estimated Labour Force}) \times 100$$

3.4 Step 4: Standard Error Estimation

Given the complex multi-stage stratified design, standard errors cannot be computed using simple random sampling formulas. Methods used:

- Taylor linearisation (approximate variance estimation)
- Jackknife replication
- Bootstrap methods

Published estimates include standard errors and coefficients of variation (CV) to indicate reliability.

3.5 Step 5: Disaggregation

Indicators are published by:

- Sex (male/female)
- Sector (rural/urban)
- Age group (15-29, 30-59, 60+)
- State/UT (for annual estimates, sample permitting)
- Industry (NIC 2008 sections)
- Occupation (NCO 2015 major groups)
- Employment status (self-employed/regular/casual)

4. Administrative Data Sources for Employment Estimation

PLFS household survey data is supplemented and validated against administrative data:

****EPFO (Employees' Provident Fund Organisation)**:** Monthly payroll data for organised-sector formal employment. EPFO net new subscribers is used as a high-frequency indicator of formal employment creation.

****ESIC (Employees' State Insurance Corporation)**:** Covers workers in establishments with 10+ employees in notified sectors. Net new ESIC registrations indicate formal employment trends.

****NPS (National Pension System)**:** Subscriber data from PFRDA for government and private sector formal employees.

****Payroll-based employment estimates**:** MoSPI compiles quarterly employment estimates for 8 key sectors using enterprise-level administrative data. This provides a supply-side (employer) perspective complementing the demand-side PLFS.

****CMIE (Centre for Monitoring Indian Economy)**:** Consumer Pyramids Household Survey provides monthly labour market data; used for comparison and validation but not official estimates.

5. Interpretation and Common Pitfalls

****Pitfall 1 - Comparing across reference periods**:** LFPR based on UPS " LFPR based on CDS. Always state the reference period clearly.

****Pitfall 2 - Female LFPR in India**:** India's female LFPR appears low by international standards but household work and subsistence activity are excluded. Using UPSS (principal + subsidiary) gives higher female WPR than UPS alone.

****Pitfall 3 - Low unemployment " full employment**:** In the absence of social protection, unemployed; they accept low-quality, low-wage casual work. This is reflected in high CDS unemployment (capturing underemployment).

****Pitfall 4 - Urban vs. rural structural differences**:** Urban labour markets are more formalised with higher WPR among educated youth; rural labour markets have higher agricultural self-employment. Conflating the two distorts policy conclusions.

****Pitfall 5 - CV-based reliability**:** Sub-group estimates (e.g., by state and industry simultaneously) may have high CVs due to small cell sizes. Treat such estimates with caution and report standard errors alongside point estimates.

6. Practical Example: Computing State-level LFPR

Hypothetical State Y, Annual PLFS data, UPS reference period, persons aged 15+:

Category	Persons (weighted, '000s)
Own-account workers	8,500
Employers	400
Contributing family workers	3,200
Regular/salaried employees	6,100
Casual wage labourers	4,800
****Total Employed****	****23,000****
Unemployed (seeking + available)	1,800
****Labour Force****	****24,800****
Out of Labour Force (students, domestic duties, etc.)	15,200
****Population 15+****	****40,000****

LFPR = $(24,800 / 40,000) \times 100 = 62.0\%$

WPR = $(23,000 / 40,000) \times 100 = 57.5\%$

Unemployment Rate = $(1,800 / 24,800) \times 100 = 7.3\%$

7. Data Quality Considerations

- ****Proxy reporting****: In PLFS, household head or one informed member may report for all household members, causing classification errors for women and youth.
- ****Misclassification of contributing family workers****: Workers in family enterprises may be reported as homemakers if they don't perceive their work as "employment."
- ****Seasonal bias****: If fieldwork is concentrated in a particular season, annual estimates may not be representative of the full year.
- ****Response burden****: Quarterly follow-up for panel households can lead to respondent fatigue and attrition.
- ****Comparability over time****: Changes in survey design, questionnaire, or processing can break time series. Document methodological changes in metadata.

8. Key Learning Points

1. Employment-unemployment indicators are estimated from PLFS using weighted survey data with complex design adjustments.
2. The three reference periods (UPS, CWS, CDS) produce different indicator values; CDS gives the broadest measure of underemployment.
3. ILO/ICLS 2013 definitions require all three conditions (without work, available, seeking) for unemployment classification.
4. Standard errors must be computed using methods appropriate for complex survey designs (Taylor linearisation, jackknife, bootstrap).
5. Administrative data (EPFO, ESIC, NPS) supplement survey estimates for high-frequency formal employment monitoring.
6. Low unemployment rates in India often reflect disguised underemployment rather than full employment.

9. Learning Outcomes

On completing this material, you should be able to:

- Apply ILO/ICLS 2013 definitions to classify persons into employment status categories.
- Explain the step-by-step methodology for estimating LFPR, WPR, and unemployment rate from PLFS data.
- Compute employment indicators from survey data using appropriate weights.
- Identify and account for complex survey design in standard error estimation.
- Describe administrative data sources used to validate and supplement PLFS-based estimates.
- Avoid common pitfalls in interpreting India's employment-unemployment statistics.